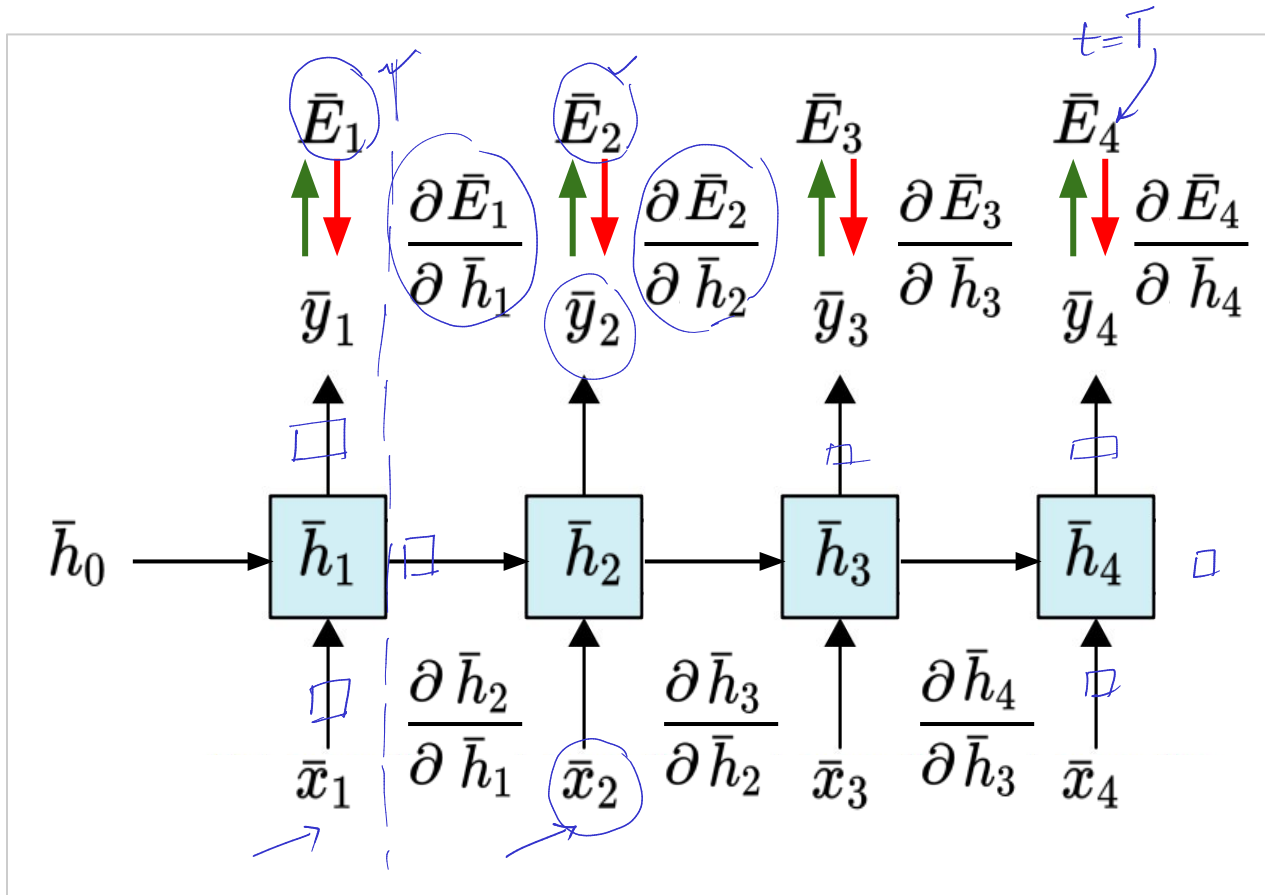


Training a recurrent neural network



skip MATN

$$\frac{\partial E_t}{\partial W_{hh}} = \sum_{1 \leq k \leq t} \frac{\partial E_t}{\partial h_t} \frac{\partial h_t}{\partial h_k} \frac{\partial h_k}{\partial W_{hh}}$$

$$\frac{\partial h_t}{\partial h_k} = \prod_{t \geq i > k} \left(\frac{\partial h_i}{\partial h_{i-1}} \right) = \prod_{t \geq i > k} W_{hh}^T \text{diag}[\phi'(h_{i-1})]$$

$\rightarrow 0$

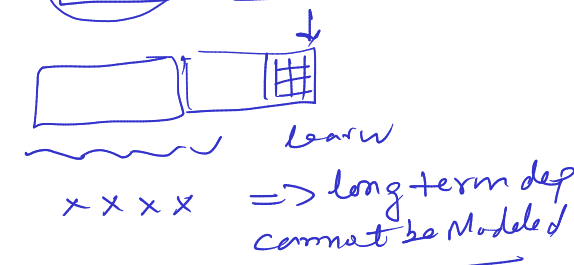
W_{hh}

- Repeated matrix multiplications leads to **vanishing and exploding gradients**
- Long-term interactions are given exponentially smaller weights than small-term interactions.

$$\nabla_t = \underbrace{\nabla_{st}}_{\text{skip}} + \nabla_{bt}$$

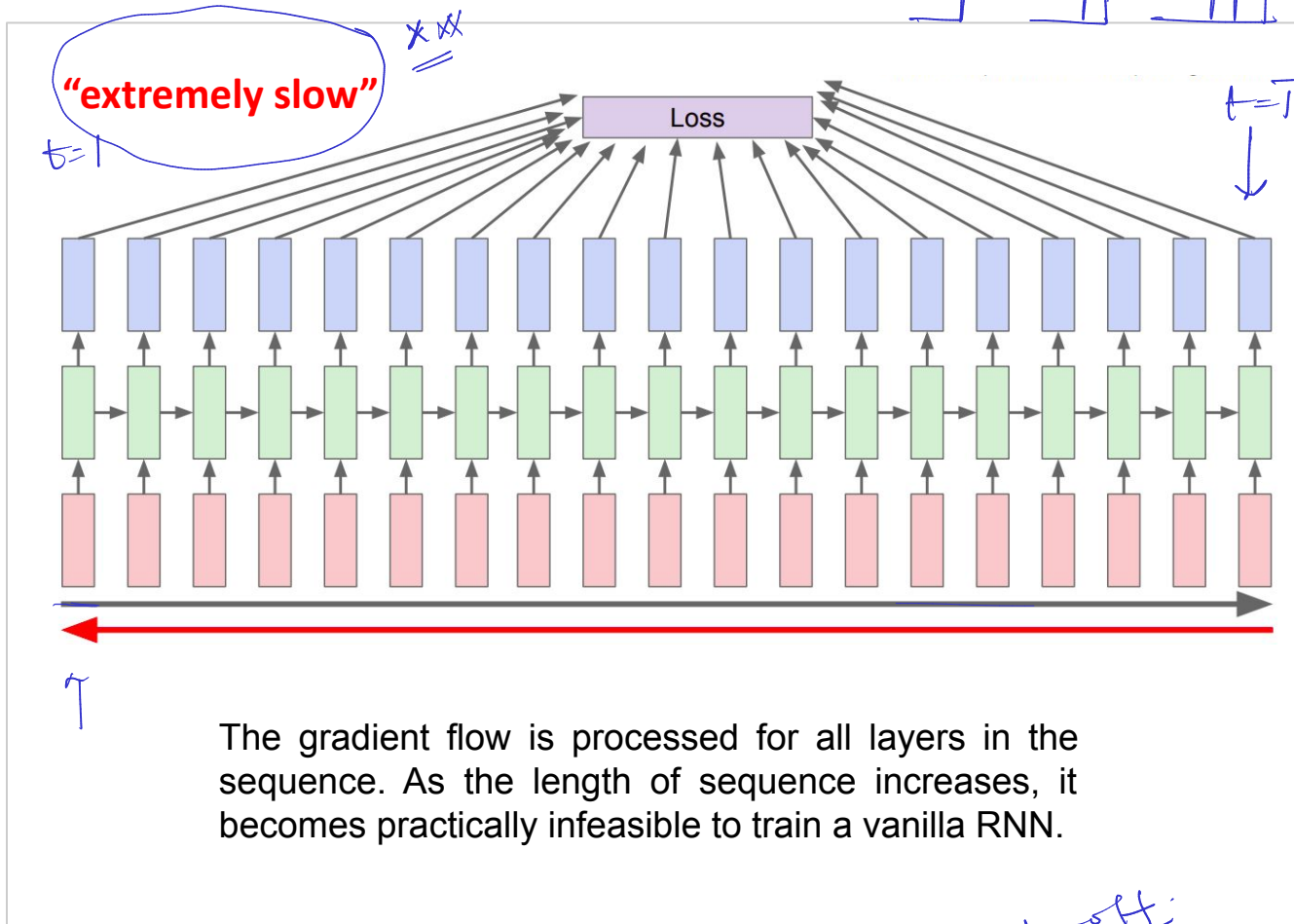
Backpropagation through Time (BPTT)

Central drawback



Backpropagation through Time (BPTT)

Seq length $H \uparrow$



Trade-off:

